

Ali Ghasemzadeh

Sharif University of Technology (SUT) Tehran, Iran

lyqasmzadh72@email.com | github.com/Alighasemzadeh83 | linkedin.com/in/ali-ghasemzadeh-bb6904266

Profile

Fourth-year Computer Engineering student at Sharif University of Technology, with a minor in Mathematics. My research spans **optimization** (convex, non-convex, and evolutionary methods), **machine learning**, and **computer vision**, including adversarial training, domain adaptation, and LLM-based reasoning. Supported by a strong mathematical background and extensive self-study, I have built solid foundations in optimization and learning theory.

I complement this grounding with substantial **practical expertise**: implementing and extending deep-learning and vision models in PyTorch, designing experiments, and planning ablations. Through contributing to two surveys and two research papers, I have gained experience in **literature synthesis**, **paper writing**, **experimental methodology**, and structuring large-scale reviews.

As the **lead researcher within the team** in several projects, I have organized the team, assigned tasks based on members' strengths, contributed to idea development, and carried out a major share of the implementation and experimentation across all stages of the work.

Education

Sharif University of Technology (SUT)

B.Sc. in Computer Engineering | Minor: Mathematics

2022 – Present (Exp. 2027)

GPA: 18.8/20 (3.83/4.00, Top 25%)

National University Entrance Exam (Konkur)

Nationwide Rank: 69 / 150,000 (Top 0.05%)

2022

National Organization for Development of Exceptional Talents (NODET)

High School Diploma in Mathematics and Physics

2019 – 2022

GPA: 19.5/20 (3.95/4.00)

Publications

- **Toward Accurate RUL and SOH Estimation Using Reinforced Graph-based PINNs** (under review, *Journal of Manufacturing Systems*) Role: Co-author contributed to conceptual design, PINN implementation, and RL-based weighting module. [arXiv](#)
- **Learning Under Joint Domain Evolution: Progressive Multi-Source Domain Adaptation** (under review, *CVPR 2026*) Role: Lead author originated the idea, developed temporal-alignment framework, implemented full adaptation pipeline, and led writing.

Research & Experience

- **Learning Under Joint Domain Evolution: Progressive Multi-Source Domain Adaptation** (under review at *CVPR 2026*) Jan–Nov 2025
Advisors: Dr. Ali Diba (HBKU), Dr. Farzan Farnia (CUHK), Dr. Babak Khalaj (Sharif)
Role: Lead author & main implementer
 - Developed the primary algorithmic framework and main PyTorch implementation.
 - Designed the temporal-alignment strategy for evolving source/target distributions.
 - Conducted extensive experiments and ablations for model evaluation.
 - Led writing and coordinated collaboration.
- **Remaining Useful Life (RUL) Prediction via PINNs and Reinforcement Learning** (under review at *Journal of Manufacturing Systems*) Jan–Jun 2025
Advisors: Dr. Mohammad Behshad Shafii (Sharif), Dr. MohamadAli Bijarchi (Sharif)
Role: Co-author & main implementer
 - Designed and implemented the main PINN modules and the RL-based dynamic weighting mechanism.
 - Ran experiments improving RUL/SOH estimation and performed ablation analysis.
 - Co-authored the manuscript.
- **Automated Boundary Update for White-Box and Black-Box Optimization** Mar 2025 – Present
Advisor: Prof. Amir H. Gandomi (UTS)
Role: Algorithm designer & implementer
 - Developed automated boundary-update strategies generalizing classical white-box methods.
 - Integrated surrogate-assisted modeling for constraint and objective evaluation.
 - Implemented core algorithmic components and defined the decision framework.
- **LLM-Driven Metaheuristic Code Generation** Jun 2025 – Present
Advisor: Prof. Seyedali Mirjalili (Torrens University Australia)
Role: Survey lead
 - Designed the survey structure and taxonomy for LLM-generated metaheuristics.
 - Reviewed and selected key papers; authored core sections.
 - Coordinated a small research team.

- **LLM and Reasoning** Jun 2025 – Present
Advisors: Dr. Erik Camberia (MIT, NTU)
Role: Literature review
 – Conducting literature review on LLMs and reasoning.
- **Constraint-Generalized Nonlinear Benchmark Generator (GNBG)** Oct 2025 – Present
Advisor: Prof. Amir H. Gandomi (UTS)
Role: Method & implementation contributor
 – Extended GNBG to generate constrained optimization tasks with tunable sparsity, feasibility ratio, and scaling.
 – Implemented key components and conducted related-work analysis.
- **Medical Reasoning with LLMs** Jul 2025 – Present
Advisors: Dr. Reza Azad (RWTH Aachen), Dr. Dorit Merhof (Regensburg)
Role: Co-lead of survey structure & writing
 – Designed the overall structure of the survey and developed a thematic taxonomy of medical reasoning tasks.
 – Performed large-scale literature selection and annotation across reasoning scenarios and evaluation paradigms.
 – Contributed extensively to writing, synthesis, and methodological framing of the survey.
- **Semidefinite Programming & Robust Optimization** Oct 2025 – Present
Advisor: Dr. Peyman Mohajerin Esfahani (University of Toronto)
Role: Theoretical research trainee
 – Conducting in-depth study of SDP.
 – Reviewing foundational books and recent research to prepare for methodological development.
- **Enhancing DiscoCLIP for Compositionality and Quantum-Circuit Modeling** Sep 2025 – Present
Advisors: Prof. Mehrnoosh Sadrzadeh (UCL), Dr. Mina Abbaszadeh (UCL), Dr. Kin Ian Lo (UCL)
Role: Methodology & implementation contributor
 – Improving the DiscoCLIP image encoder to strengthen compositional representations.
 – Integrating image-encoding mechanisms capable of modeling quantum-circuit structures.
 – Conducting literature review, implementing encoder variants, and running empirical evaluations.

Honors & Awards

- **Top 2 / 70** students in Convex Optimization Fall 2025
- **Top 1 / 10** students in Estimation Theory Spring 2025
- **Top 0.05%** in National University Entrance Exam (Konkur) 2022
- **Top 25%** of cohort based on GPA Present

Selected Coursework

- **Advanced:** Convex Optimization, Generative Models, Reinforcement Learning, Estimation Theory, Stochastic Processes, Machine Learning.
- **Foundational:** AI, Linear Algebra, Probability & Statistics, Data Structures, Automata Theory.

Self-Study & Certifications

- Online Learning & Bandits (Sharif, Kasra Alishahi)
- Deep Learning (Sharif, Mahdiah Soleymani)
- Generative Models (Stanford, Ermon)
- Convex Optimization (Stanford, Boyd)
- Game Theory (Sharif, Kasra Alishahi)

Teaching Assistantships (Practice Design)

- ML Theory (Fall 2025)
- ML (Spring 2025)
- Probability & Statistics (Spring & Fall 2024)
- Linear Algebra (Fall 2024)
- Automata Theory (Fall 2024)
- Fundamentals of Programming (Fall 2023)

Technical & Research Skills

- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, diffusion models, GANs, RL.
- **Optimization & Theory:** Convex/non-convex optimization, estimation theory, probability, stochastic processes, linear algebra.
- **Software:** Python, Git, Linux; reproducible research workflows.
- **Research Tools:** L^AT_EX, scientific writing, paper summarization, mentoring.
- **Languages:** Persian (native), English (fluent).

Selected Repositories

- **QSM-for-Games** Score-matching generative models for multi-domain game datasets. [GitHub](#)
- **Pix2Pix-from-Scratch** Paired image-to-image GAN implemented in PyTorch. [GitHub](#)
- **U-Net Semantic Segmentation** Full PyTorch segmentation pipeline using U-Net. [GitHub](#)
- **Reinforcement Learning (A2C)** ActorCritic implementation for control tasks. [GitHub](#)